

COVER LETTER

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Hiring Committee

College of Engineering — Robotics Department

University of Michigan

Ann Arbor, MI

Dear Hiring Committee,

I am writing to express interest in the Applications Programmer Associate (Job ID 270067) in the Robotics Department. With an M.S. in Computer and Information Science, ~2 years of professional software engineering experience, and a part-time Graduate Assistantship supporting academic technology, I bring a practical blend of web application development, application support, database work, and higher-education experience that directly matches the responsibilities and culture described in the posting.

At BloomTech Solutions (Sep 2023–Sep 2025) I designed, implemented, and maintained web and cloud-based applications using Python, JavaScript, and SQL, focusing on reliability and secure data exchange. I built and debugged RESTful APIs, wrote optimized SQL scripts for backend data workflows, and automated builds and deployments with Jenkins and Docker. I routinely collaborated with support teams to triage incidents, document resolutions, and track tickets using ServiceNow, applying an ITIL-oriented incident workflow to ensure timely closures. My teamwork in Agile sprints and consistent use of Git/GitHub for version control ensured iterative, well-documented releases—skills I will bring to Michigan when supporting departmental applications and research tools.

My part-time role as a Graduate Assistant (Jan 2024–Dec 2024) at Western Illinois University gave me direct experience with the operational needs of academic environments. There I supported faculty and students by troubleshooting lab software and instructional tools, authoring technical documentation, executing SQL queries to assist faculty reporting, and participating in requirement-gathering sessions to convert instructional needs into practical tools. This experience taught me how to maintain classroom and research workflows, handle sensitive academic data responsibly, and communicate technical solutions clearly to non-technical stakeholders—capabilities that align with the Robotics Department's need for application support and user-focused development.

I also bring experience supporting cloud environments (Azure, AWS) and applying security and compliance practices during development and maintenance—important when working with research data and university systems. In addition to hands-on technical skills, I contribute strong testing and QA practices (unit testing, test case execution, defect reporting) and an eagerness to learn domain-specific tools the department uses. I am comfortable working in a hybrid environment and collaborating closely with faculty, staff, and students to deliver tools that improve productivity and research outcomes.

I am excited by the opportunity to support the Robotics Department's applications and help turn user needs into secure, reliable tools that enable teaching and research. Thank you for considering my application. I have attached my resume with this letter and would welcome the chance to discuss how my technical background and higher-education experience can contribute to your team.

Sincerely,

Navya Koduri

PROFESSIONAL SUMMARY

Detail-oriented Applications Programmer and M.S. in Computer and Information Science graduate with hands-on experience in web development, database design, and application support. Proficient in JavaScript, Python, and SQL, with a strong foundation in debugging, testing, and collaborating with cross-functional teams. Skilled at translating user needs into efficient, secure applications and tools. Eager to contribute technical expertise to support the University of Michigan's mission of innovation and excellence in robotics education.

EDUCATION

Master of Science in Computer and Information Science

Western Illinois University - Macomb, IL

December 2024

GPA:3.81

TECHNICAL SKILLS

- **Programming:** JavaScript, Python, Java, SQL, PowerShell
- **Web Development:** HTML, CSS, REST APIs, Flask, Node.js
- **Databases:** MySQL, SQL Server, Oracle
- **Testing & QA:** Unit testing, debugging, test case design, defect tracking
- **Tools & Technologies:** Git/GitHub, Jenkins, Docker, ServiceNow, VS Code
- **Cloud Platforms:** Microsoft Azure, AWS
- **Operating Systems:** Windows, Linux
- **Networking & Systems:** Basic TCP/IP, virtualization, troubleshooting
- **Methodologies:** Agile, Scrum, CI/CD, ITIL Incident Management
- **Soft Skills:** Requirement gathering, teamwork, documentation, communication, user support

WORK EXPERIENCE

Software Engineer –BloomTech Solutions– Remote, TX

| September 2023 – September 2025

- Developed and maintained web-based applications using Python, JavaScript, and SQL, ensuring high reliability, usability, and alignment with project requirements.
- Designed and debugged RESTful APIs to enable secure data exchange between systems and improve application interoperability.
- Supported database-driven applications, writing optimized SQL queries and managing data updates to maintain accuracy and performance.
- Collaborated with development and support teams to analyze, troubleshoot, and document application issues, improving response time and user satisfaction.
- Participated in requirements analysis and user feedback sessions to translate functional needs into effective and secure software solutions.
- Conducted testing, validation, and debugging to ensure application integrity, minimize defects, and maintain coding compliance.
- Utilized Jenkins and Docker for automating builds and deployments, streamlining version releases and minimizing manual intervention.
- Assisted in managing cloud environments (Azure, AWS), supporting updates, and applying security best practices to maintain compliance and data protection.
- Maintained documentation and version control through Git/GitHub and contributed actively to Agile sprints, promoting collaboration and iterative improvement.

Graduate Assistant– Western Illinois University –Part-Time, Macomb, IL

| January 2024 – December 2024

- Assisted in developing and maintaining small-scale web applications and automation scripts used in academic labs, applying Python and JavaScript for workflow efficiency.
- Supported faculty and students by troubleshooting technical issues, documenting fixes, and ensuring minimal downtime during coursework and research activities.
- Wrote and executed SQL queries to manage and retrieve departmental data, supporting faculty reporting and lab system updates.
- Participated in requirement gathering and analysis, collaborating with professors to identify and translate instructional needs into practical technical tools.
- Followed established security and compliance protocols for managing user access and sensitive student data, ensuring adherence to university IT standards.

- Contributed to testing and quality assurance of academic tools, helping to detect and resolve defects prior to deployment.
- Collaborated with IT staff and teaching faculty to streamline academic workflows, improving data accessibility and operational efficiency.
- Gained insight into higher education system management, strengthening ability to support applications and workflows in academic and research environments.

ACADEMIC PROJECTS

- ❖ **Database System for Robonix International Inc.** | **October 2024 – December 2024**
 - Designed and implemented a fully normalized database in BCNF to support sales, leasing, maintenance, customer feedback, and financial tracking for a global robotics company.
 - Built and optimized complex SQL queries for revenue tracking, lease management, and operational reporting.
 - Delivered a robust solution that improved data integrity, eliminated redundancy, and provided a scalable database platform for multi-role operations.
 - Developed a Java-based GUI with JDBC for dynamic role-based access, enabling sales, management, and technicians to interact with tailored views of company data.
 - Populated the system with realistic test data using Mockaroo and scripts, ensuring high-quality data consistency across modules.
- ❖ **Voice-Based Email System for Visually Impaired Users** | **March 2024 – May 2024**
 - Designed and implemented a Python-based email system that enabled visually impaired users to send and receive emails entirely through voice commands.
 - Integrated Speech-to-Text (DeepSpeech) and Text-to-Speech (gTTS, pyttsx3) with SMTP, ensuring hands-free and accessible communication.
 - Conducted usability testing to validate functionality and improved recognition accuracy in noisy environments through pre-processing techniques.
 - Enhanced system responsiveness by minimizing delays in real-time voice interactions and adapting to microphone input Variability.
- ❖ **Azure Cloud Resource Manager Dashboard** | **September 2023 – December 2023**
 - Designed simple visualizations to display resource status and usage information for easy monitoring.
 - Implemented user login and access control with Azure Active Directory to secure the dashboard.
 - Developed backend services to collect resource data and generate basic reports for cloud administrators.
 - Focused on building a user-friendly interface with responsive design to provide clear insights into cloud resources.
 - Practiced key concepts such as resource grouping, tagging, and API integration relevant to cloud management.

CERTIFICATIONS

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| • Oracle | – Oracle Cloud Infrastructure AI Foundations Associate | – October 2025 |
| • Oracle | – Oracle Cloud Infrastructure Foundations Associate | – October 2025 |
| • Oracle | – AI, OCI, Multicloud & Data Certification | – September 2025 |
| • Microsoft | – Azure AI Essentials, Microsoft Copilot Professional | – March 2025 |
| • Microsoft | – Career Essentials in Generative AI | – March 2025 |
| • Docker | – Docker Foundations Professional Certificate | – March 2025 |
| • ServiceNow | – IT Leadership Professional Certificate | – March 2025 |
| • LinkedIn | – Running Jenkins on AWS | – March 2025 |
| • GitHub | – Career Essentials in GitHub Professional Certificate | – March 2025 |
| • AWS APAC | – Solutions Architecture Job Simulation from Forage | – February 2025 |